

Kafka + CompletableFuture Design (Interview Summary)

In my design, I use Apache Kafka for asynchronous communication between microservices and CompletableFuture for parallel processing within a service. For example, when a Payment Service consumes an order event from Kafka, it performs fraud check, payment processing, and audit logging concurrently using CompletableFuture. Once all tasks complete, it publishes the result back to Kafka. This combination ensures both high scalability and low latency.

Key Points:

- Kafka handles communication between services (distributed async).
- CompletableFuture handles parallel execution within a service (in-memory async).
- Improves system performance and responsiveness.

Example Flow:

1. Order Service publishes event to Kafka
2. Payment Service consumes event
3. Fraud check, payment, audit run in parallel
4. Result published back to Kafka